

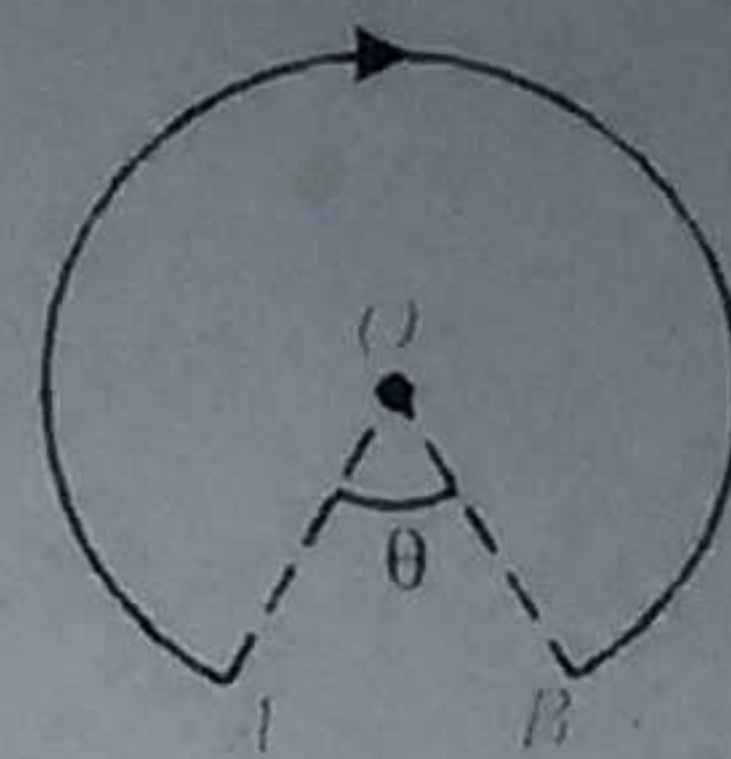
MAGNETIC EFFECTS & MAGNETIC FORCES

Type of CPP - Objective & Subjective

Level - MODERATE

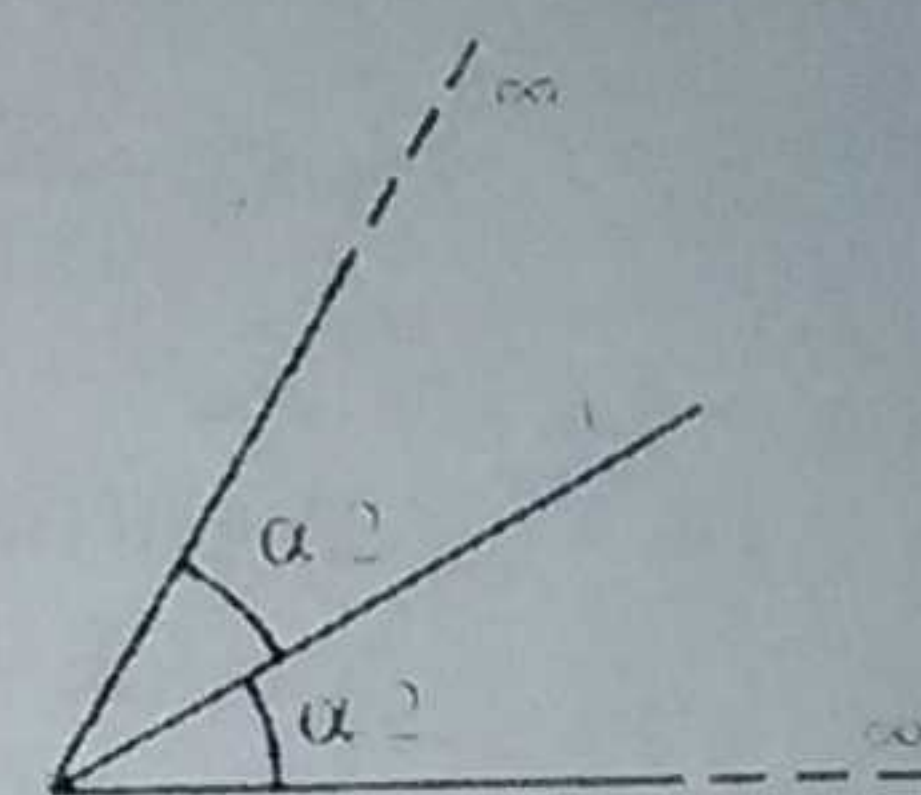
1. A current carrying wire AB of the length $2\pi R$ is turned along a circle, as shown in figure. The magnetic field at the centre O :

(A) $\frac{\mu_0 i}{2R} \left(\frac{2\pi - \theta}{2\pi} \right)^2$ (B) $\frac{\mu_0 i}{2R} \left(\frac{2\pi - \theta}{2\pi} \right)$
 (C) $\frac{\mu_0 i}{2R} (2\pi - \theta)$ (D) $\frac{\mu_0 i}{2R} (2\pi - \theta)^2$



2. Two circular coils of wire each having a radius of 4 cm and 10 turns have a common axis and are 6 cm apart. If a current of 1 A passes through each coil in the opposite direction find the magnetic induction. :
 (a) At the centre of each coil (b) At the point on the axis, midway between them

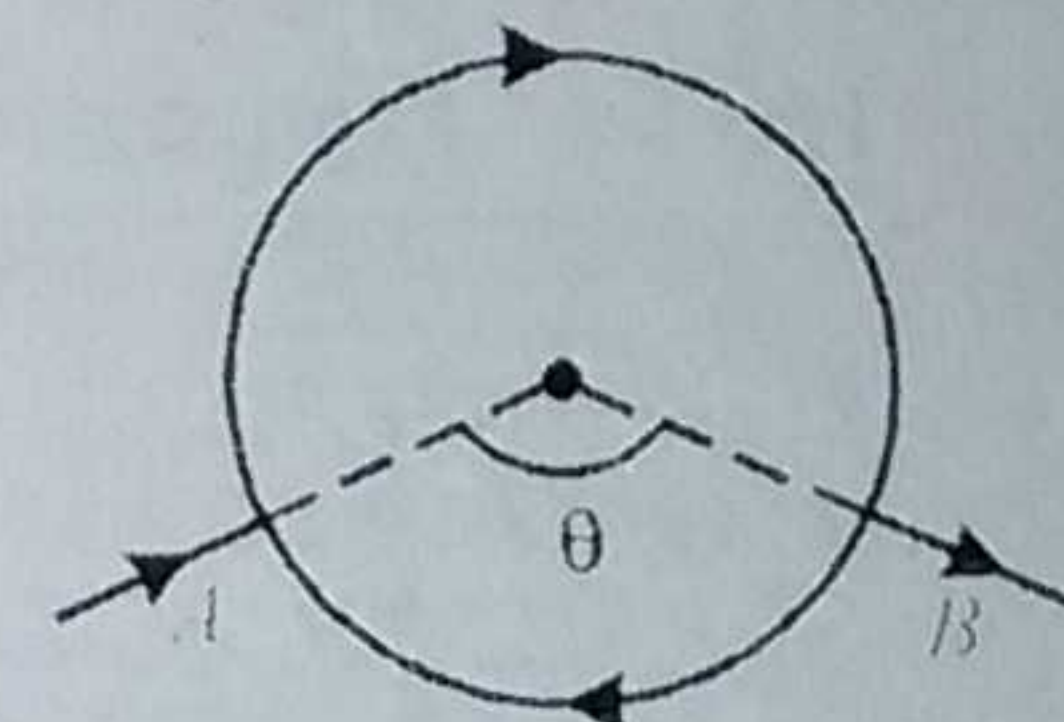
3. A long wire carrying a current i is bent to form a plane angle α . Find the magnetic field B at a point on the bisector of this angle situated at a distance x from the vertex.



4. Find the magnetic field B at the centre of a square loop of side ' a ', carrying a current i .

5. A battery is connected between two points A and B the circumference of a uniform conducting ring of radius r and resistance R . One of the arcs AB of the ring subtends an angle θ at the centre. The value of the magnetic induction at the centre due to the current in the ring is :

- (A) zero, only if $\theta = 180^\circ$
 (B) zero for all values of θ
 (C) proportional to $2(180^\circ - \theta)$
 (D) inversely proportional to r



6. A wire is wound on a long rod of material of relative permeability $\mu_r = 4000$ to make a solenoid. If the current through the wire is 5 A and number of turns per unit length is 1000 per metre, then the magnetic field inside the solenoid is :
 (A) 25.12 mT (B) 12.56 mT (C) 12.56 T (D) 25.12 T

7. A point charge is moving in a circle with constant speed. Consider the magnetic field produced by the charge at a fixed point P (not centre of the circle) on the axis of the circle :

- (A) it is constant in magnitude only (B) it is constant in direction only
 (C) it is constant in direction and magnitude both (D) it is not constant in magnitude and direction both

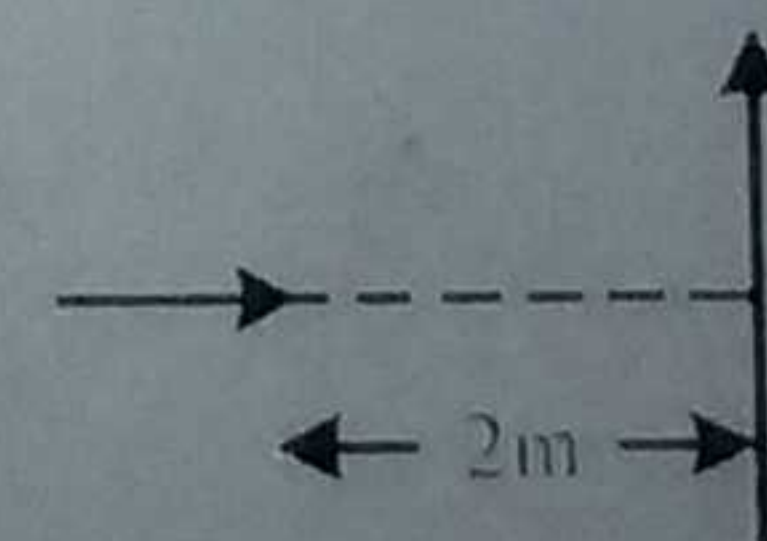
8. A copper wire having resistance 0.01 ohm in each metre is used to wind a 400 turn solenoid of radius 1.0 cm and length 20 cm. Find the emf of a battery which when connected across the solenoid will cause a magnetic field of $1.0 \times 10^{-2} T$ near the centre of the solenoid.

9. A thin but long, hollow, cylindrical tube of radius r carries a current i along its length. Find the magnitude of the magnetic field at a distance $r/2$ from the surface (a) inside the tube (b) outside the tube.

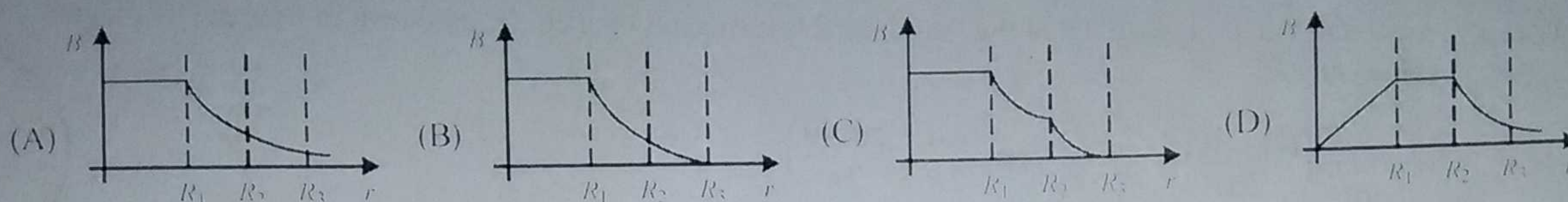
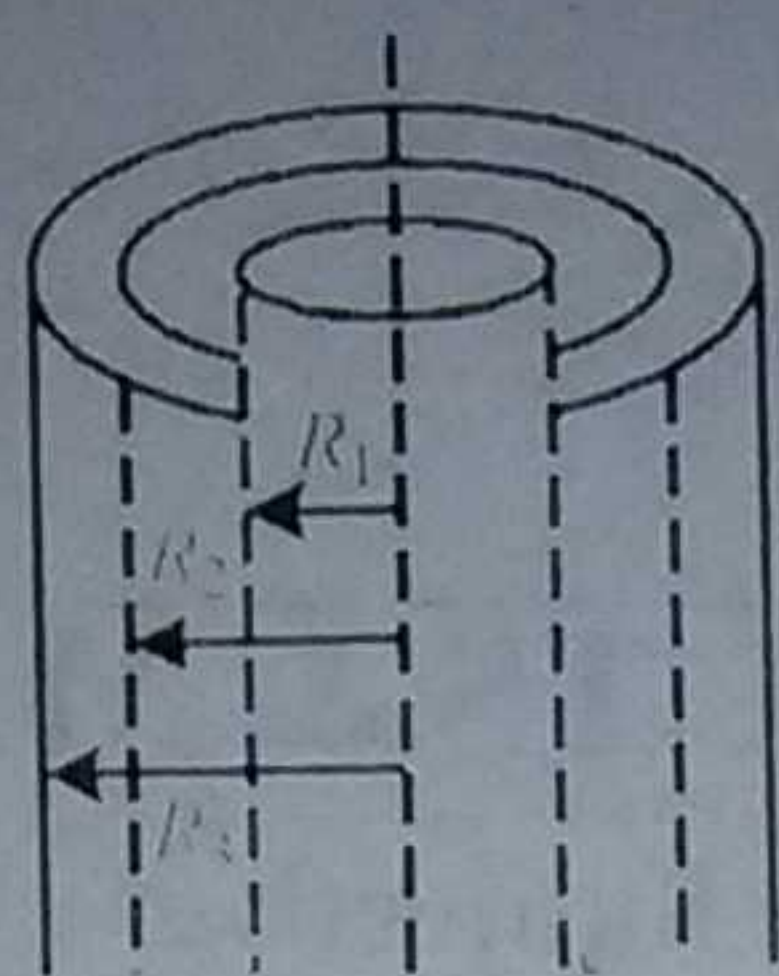
10. The magnetic field B inside a long solenoid, carrying a current of 10 A, is $3.14 \times 10^{-2} T$. Find the number of turns per unit length of the solenoid.

11. Two identical magnetic dipoles of magnetic moments $1.0 A\cdot m^2$ each, placed at a point midway between the dipole is :

(A) $5 \times 10^{-7} T$ (B) $\sqrt{5} \times 10^{-7} T$
 (C) $10^{-7} T$ (D) $2 \times 10^{-7} T$



12. A coaxial cable is made up of two conductors. The inner conductor is solid and is of radius R_1 and the outer conductor is hollow of inner radius R_2 and outer radius R_3 . The space between the conductors is filled with air. The inner and outer conductors are carrying currents of equal magnitudes and in opposite directions. Then the variation of magnetic field with distance from the axis is best plotted as :

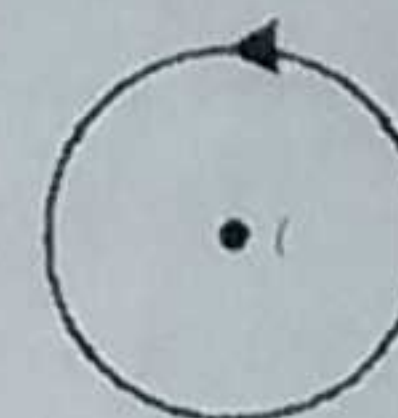


13. Axis of a solid cylinder of infinite length and radius R lies along y -axis it carries a uniformly distributed current ' i ' along $+y$ direction. Magnetic field at a point $\left(\frac{R}{2}, y, \frac{R}{2}\right)$ is :

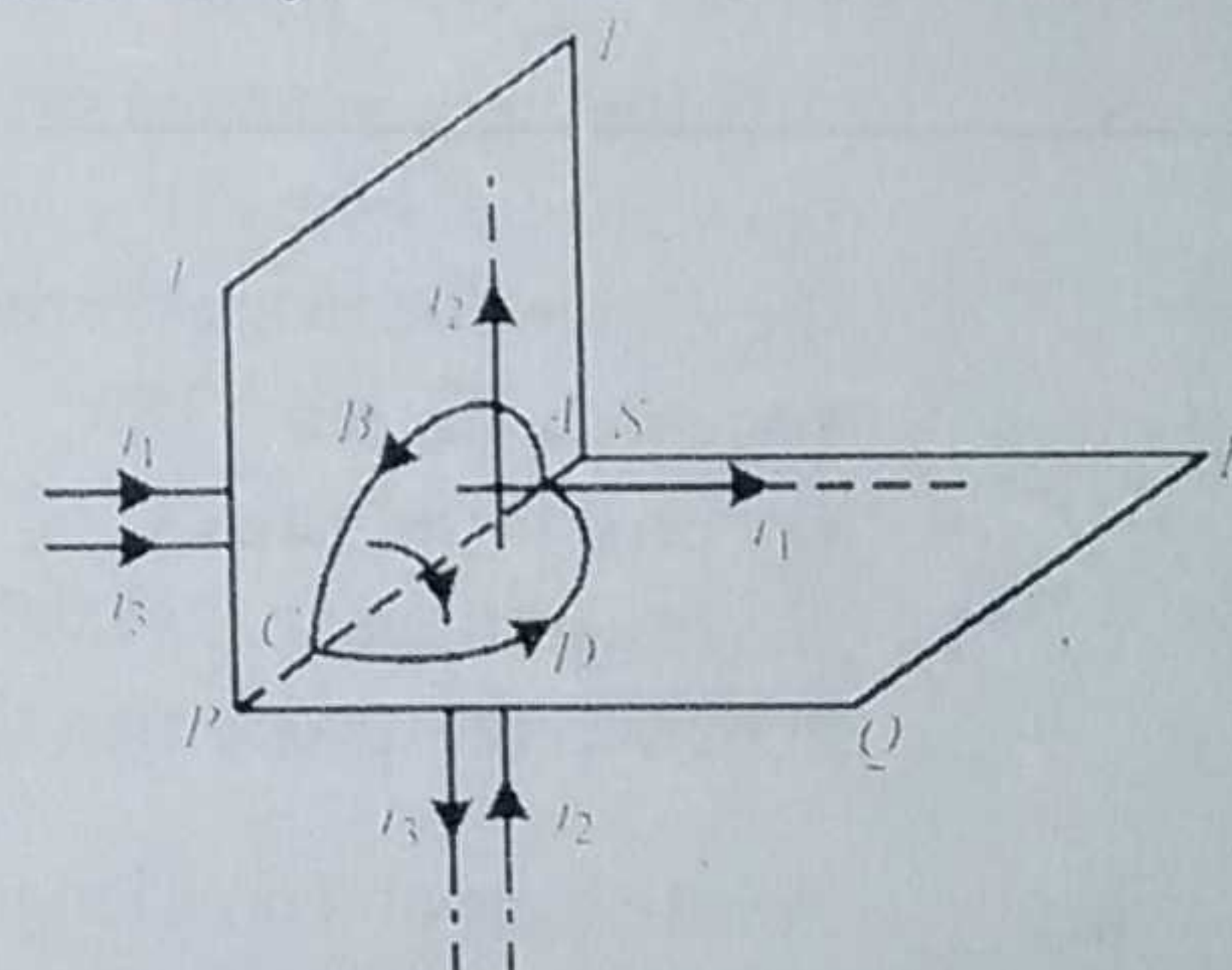
- (A) $\frac{\mu_0 i}{4\pi R}(\hat{i} - \hat{k})$ (B) $\frac{\mu_0 i}{2\pi R}(\hat{j} - \hat{k})$ (C) $\frac{\mu_0 i}{4\pi R}\hat{j}$ (D) $\frac{\mu_0 i}{4\pi R}(\hat{i} + \hat{k})$

14. A particle of negative charge of magnitude ' q ' is revolving with constant speed ' v ' in a circle of radius ' R '. Find the magnetic field (magnitude and direction) at the following points :

- (i) centre of the circle (magnitude and direction)
(ii) a point on the axis and at a distance ' x ' from the centre of the ring (magnitude only). Is its direction constant all the time ?



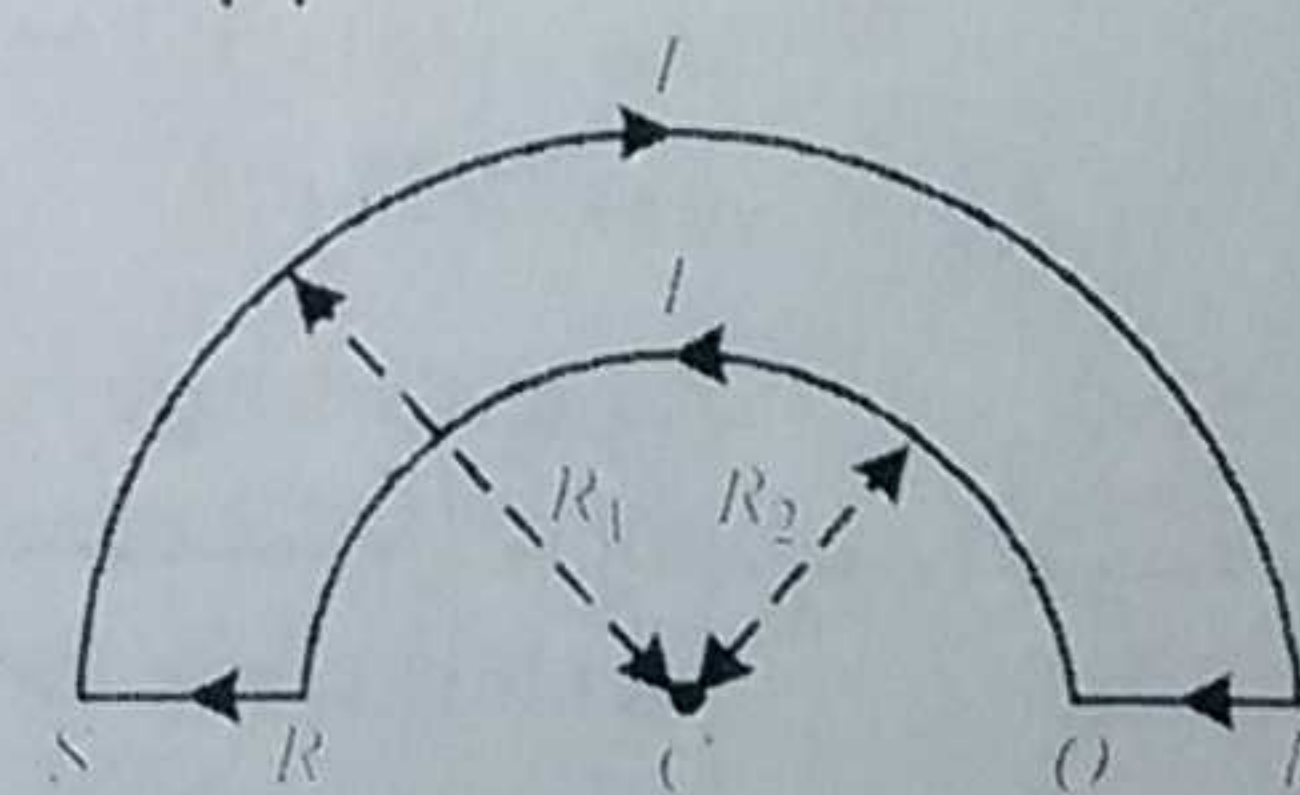
15. Figure shows an amperian path $ABCD$. Part ABC is in vertical plane $PSTU$ while part CDA is in horizontal plane $PQRS$. Direction of circulation along the path is shown by an arrow near point B and at D .



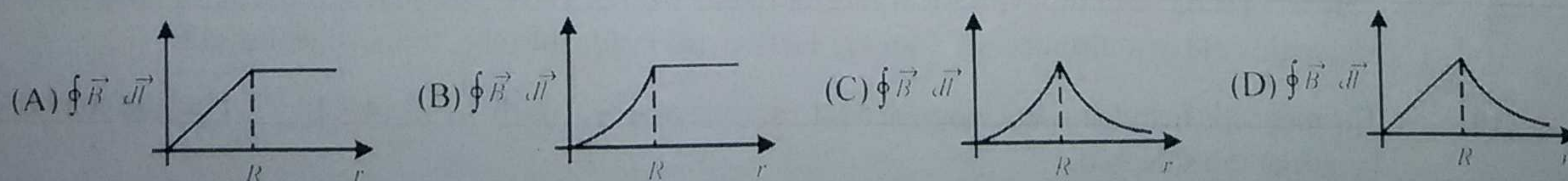
$\oint \vec{B} \cdot d\vec{l}$ for this path according to ampere's law will be :

- (A) $(i_1 - i_2 + i_3)\mu_0$ (B) $(-i_1 + i_2)\mu_0$
(C) $i_3\mu_0$ (D) $(i_1 + i_2)\mu_0$

16. Two wire loops $PQRSP$ formed by joining two semicircular wires of radii R_1 and R_2 carries a current I as shown in figure. The magnitude of the magnetic induction at the centre C is

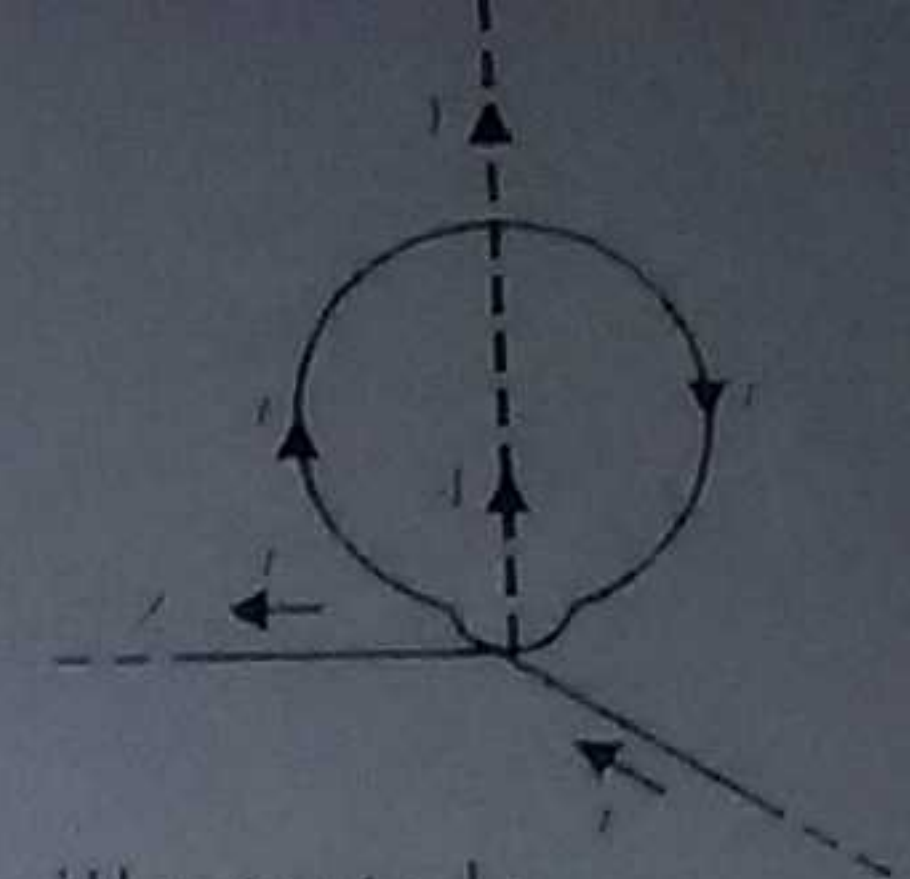


17. A cylindrical wire of radius R is carrying current i uniformly distributed over its cross-section. If a circular loop of radius ' r ' is taken as amperian loop, then the variation value of $\oint \vec{B} \cdot d\vec{l}$ over this loop with radius ' r ' of loop will be best represented by :

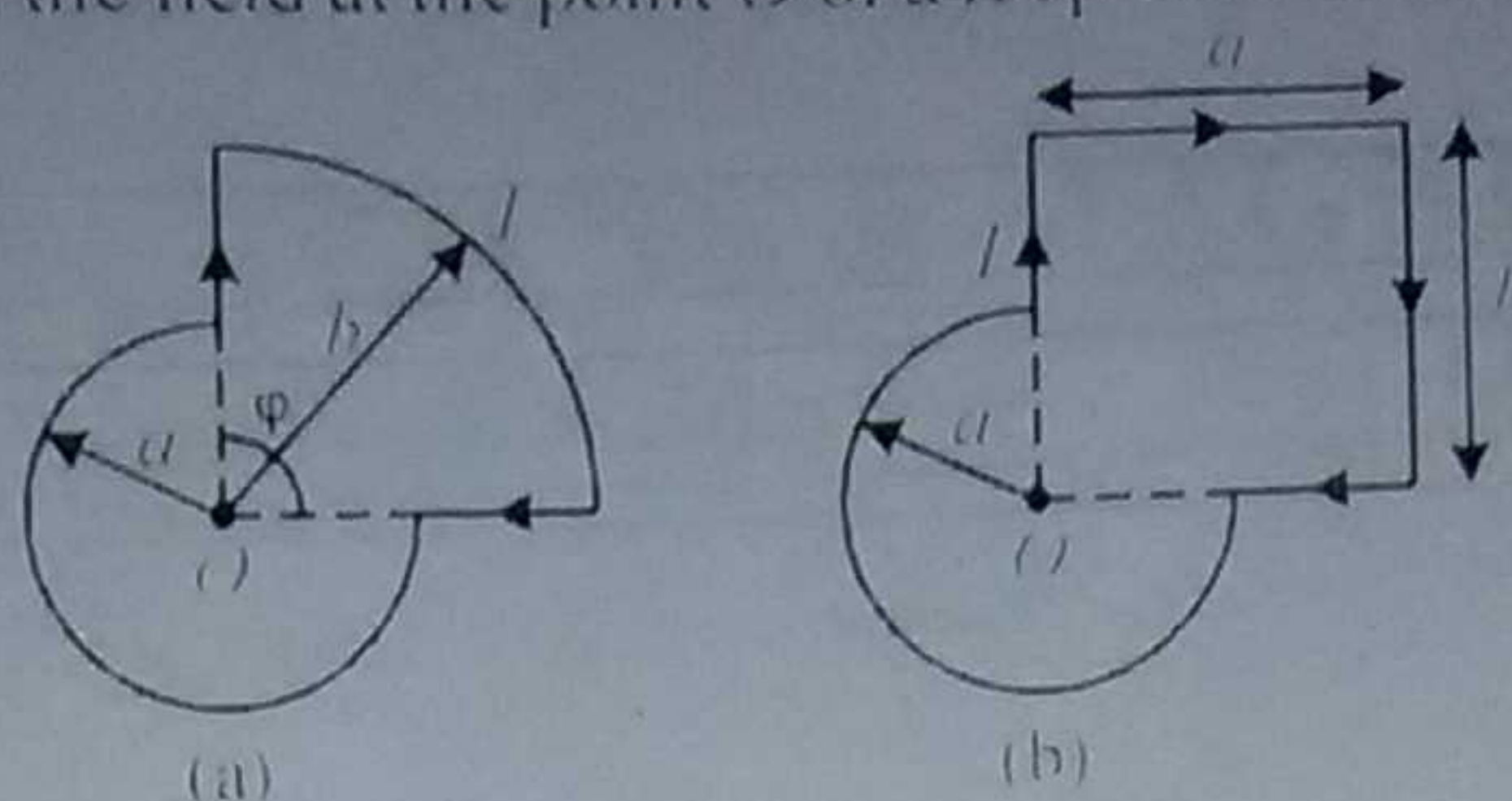


18. A magnetic needle (small magnet) is kept in a nonuniform magnetic field. It :
(A) may experience a force and torque (B) may experience a force but not a torque
(C) may experience a torque but not a force (D) will experience neither a force nor a torque

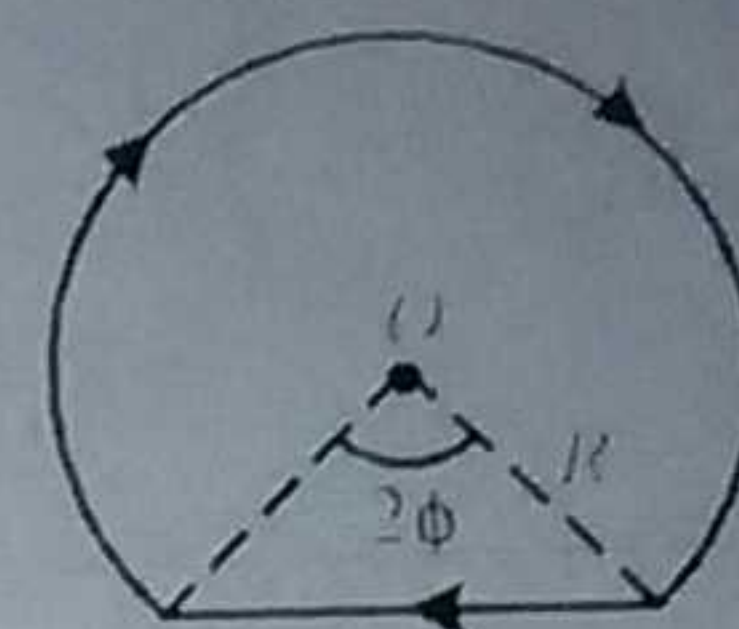
19. Find the magnitude of the magnetic induction B of a magnetic field generated by a system of thin conductors (along which a current i is flowing) at a point $A(0, R, 0)$, that is the centre of a circular conductor of radius R . The ring is in yz plane



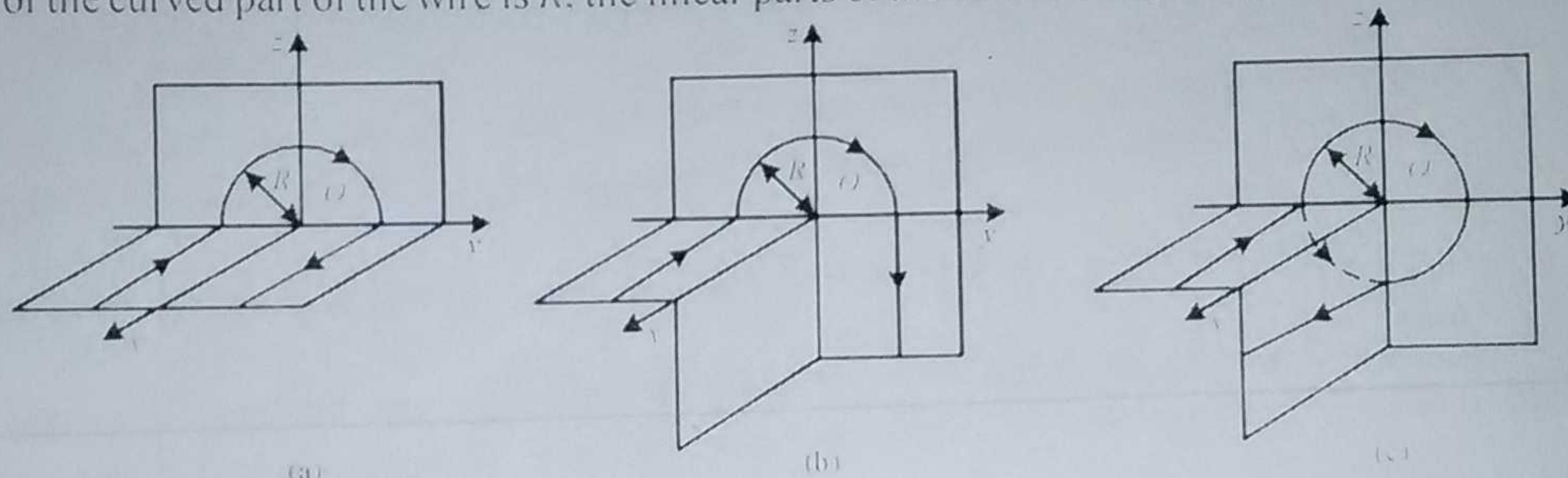
20. Find the magnetic induction of the field at the point O of a loop with current I , whose shape is illustrated :



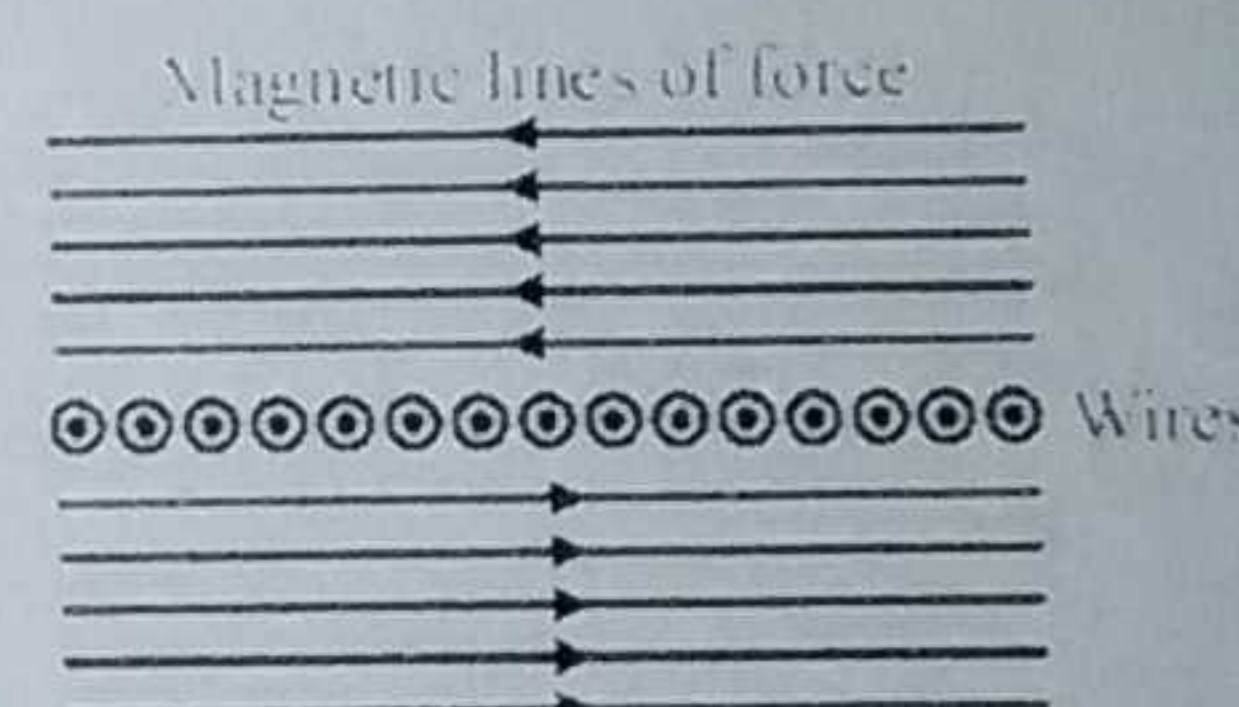
- (a) In figure (a) the radii a and b , as well as the angle ϕ are known.
 (b) In figure (b), the radius a and the side b are known.
 (c) A current $I = 5.0$ A flows along a thin wire shaped as shown in figure. The radius of a curved part of the wire is equal to $R = 120$ mm, the angle $2\phi = 90^\circ$. Find the magnetic induction of the field at the point O .



21. Find the magnetic induction at the point O if the wire carrying a current I has the shape shown in figure a, b, c . The radius of the curved part of the wire is R , the linear parts of the wire are very long.



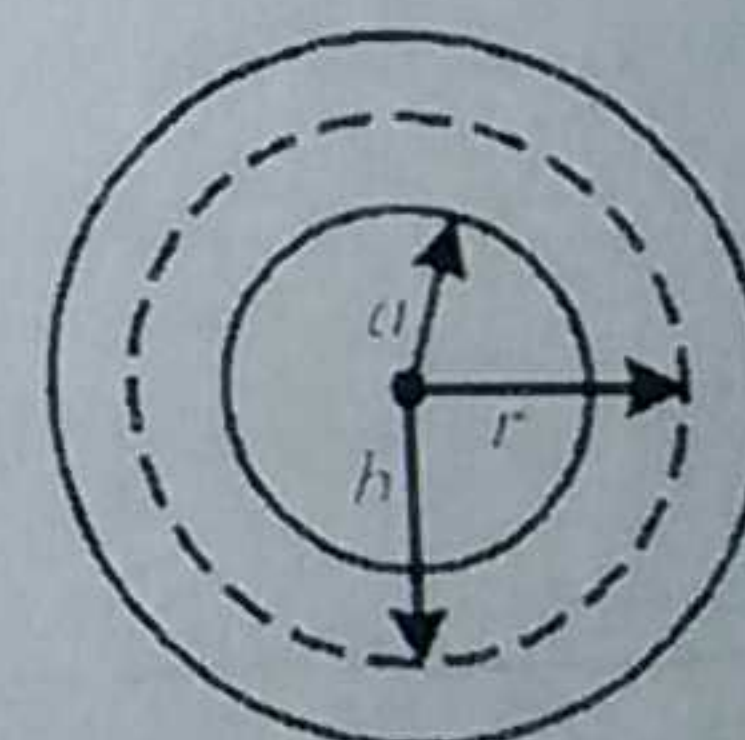
22. A conductor consists of an infinite number of adjacent wires, each infinitely long and carrying a current i . Show that the lines of B will be as represented in figure and that B for all points in front of the infinite current sheet will be given by, $B = (1/2) \mu_0 ni$, where n is the number of conductors per unit length.



23. Figure shows a cylindrical conductor of inner radius a and outer radius b which carries a current i uniformly spread over its cross section. Show that the magnetic field B for points inside the body

of the conductor (i.e. $a < r < b$) is given by, $B = \frac{\mu_0 i}{2\pi(b^2 - a^2)} \frac{r^2 - a^2}{r}$.

Check this formula for the limiting case of $a = 0$.



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ANSWER KEY

1. (A) 2. [(a) $5\pi \times 10^{-5} \left(1 - \frac{8}{13\sqrt{13}}\right) T$ (b) Zero] 3. $\left|\frac{\mu_0 i}{4\pi x} \cot \frac{\alpha}{4}\right|$ 4. $\left|\frac{2\sqrt{2}\mu_0 i}{\pi a}\right|$ 5. (B) 6. (D) 7. (A)
 8. [1 V] 9. [(a) Zero, (b) $\frac{\mu_0 i}{3\pi r}$] 10. [2500 turns/m] 11. (B) 12. (C) 13. (A) 14. [(i) $\frac{\mu_0 qv}{4\pi R^2}$, inwards
 (ii) $\frac{\mu_0 qv}{4\pi(x^2 + R^2)}$, No] 15. (D) 16. $\left[\frac{\mu_0 I}{4} \left(\frac{1}{R_1} - \frac{1}{R_2}\right)\right]$ 17. (B) 18. (A)(B)(C) 19. $|B| = \frac{\mu_0 I}{4\pi R} \sqrt{2(2\pi^2 - 2\pi + 1)}$
 20. [(a) $B = \frac{\mu_0 I}{4\pi} \left(\frac{2\pi - \phi}{a} + \frac{\phi}{b}\right)$; (b) $B = \frac{\mu_0 I}{4\pi} \left(\frac{3\pi}{2a} + \frac{\sqrt{2}}{b}\right)$; (c) $B = (\pi - \phi + \tan \phi) \mu_0 I / 2\pi R = 28 \mu T$
 21. [(a) $\vec{B} = \frac{\mu_0 I}{4\pi R} (-\pi \hat{i} - 2\hat{k})$ (b) $\vec{B} = \frac{\mu_0 I}{4\pi R} \left[-\hat{i} \left(1 + \frac{1}{\pi}\right) - \frac{\hat{k}}{\pi}\right]$ (c) $\vec{B} = \frac{\mu_0 I}{4\pi R} (-\hat{j} - \hat{k})$] 22. $|B| = \frac{\mu_0 ni}{2}$ 23. $|B| = \frac{\mu_0 i}{2\pi b^2} r$

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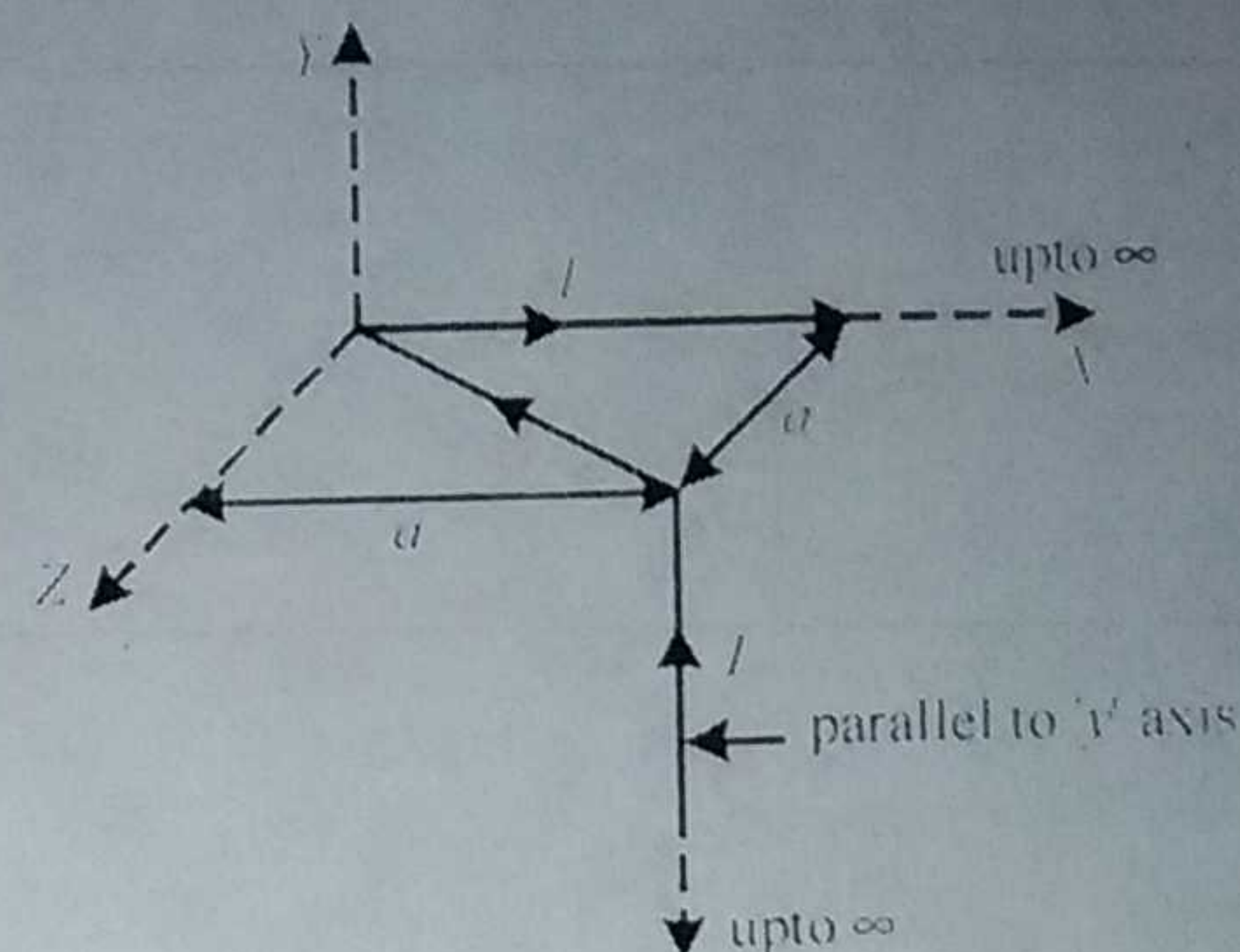
Level - MODERATE

1. The magnetic field at the origin due to a current element $i d\vec{l}$ placed at a position $\vec{r}$ is :

(A) $\frac{\mu_0 i}{4\pi} \frac{d\vec{l} \times \vec{r}}{r^3}$ (B) $-\frac{\mu_0 i}{4\pi} \frac{\vec{r} \times d\vec{l}}{r^3}$ (C) $\frac{\mu_0 i}{4\pi} \frac{\vec{r} \times d\vec{l}}{r^3}$ (D) $-\frac{\mu_0 i}{4\pi} \frac{d\vec{l} \times \vec{r}}{r^3}$

2. The magnetic field at the origin due to the current flowing in the wires is :

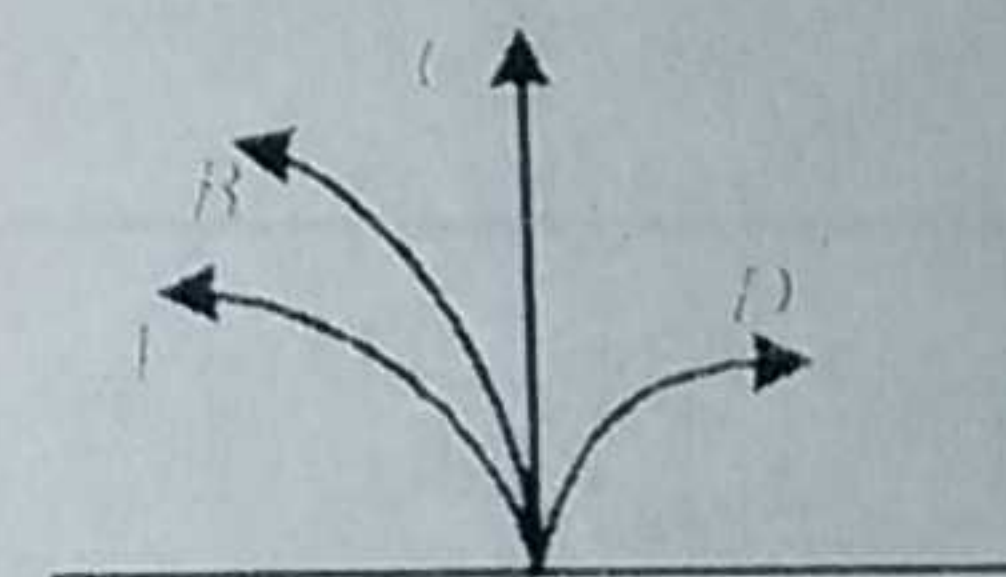
(A) $-\frac{\mu_0 I}{8\pi a} (\hat{i} + \hat{k})$
 (B) $\frac{\mu_0 I}{2\pi a} (\hat{i} + \hat{k})$
 (C) $\frac{\mu_0 I}{8\pi a} (-\hat{i} + \hat{k})$ (D) $\frac{\mu_0 I}{4\pi a\sqrt{2}} (-\hat{i} - \hat{k})$



3. Consider three quantities $x = E/B$, $y = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ and $z = \frac{l}{CR}$. Here, l is the length of a wire, C is a :
 (A) x, y have the same dimensions. (B) y, z have the same dimensions.
 (C) z, x have the same dimensions. (D) None of the three pairs have the same dimensions

4. A charged particle is accelerated through a potential difference of 12 kV and acquires a speed of 1.0×10^6 m/s. It is then injected perpendicularly into a magnetic field of strength 0.2 T. Find the radius of the circle described by it.

5. A neutron, a proton, an electron and an α -particle enters a uniform magnetic field with equal velocities. The field is directed along the inward normal to the plane of the paper. Which of these tracks followed are by electron and α -particle.



6. In the formula $X = 3YZ^2$, the quantities X and Z have the dimensions of capacitance and magnetic induction respectively. The dimensions of Y in the MKSQ system are

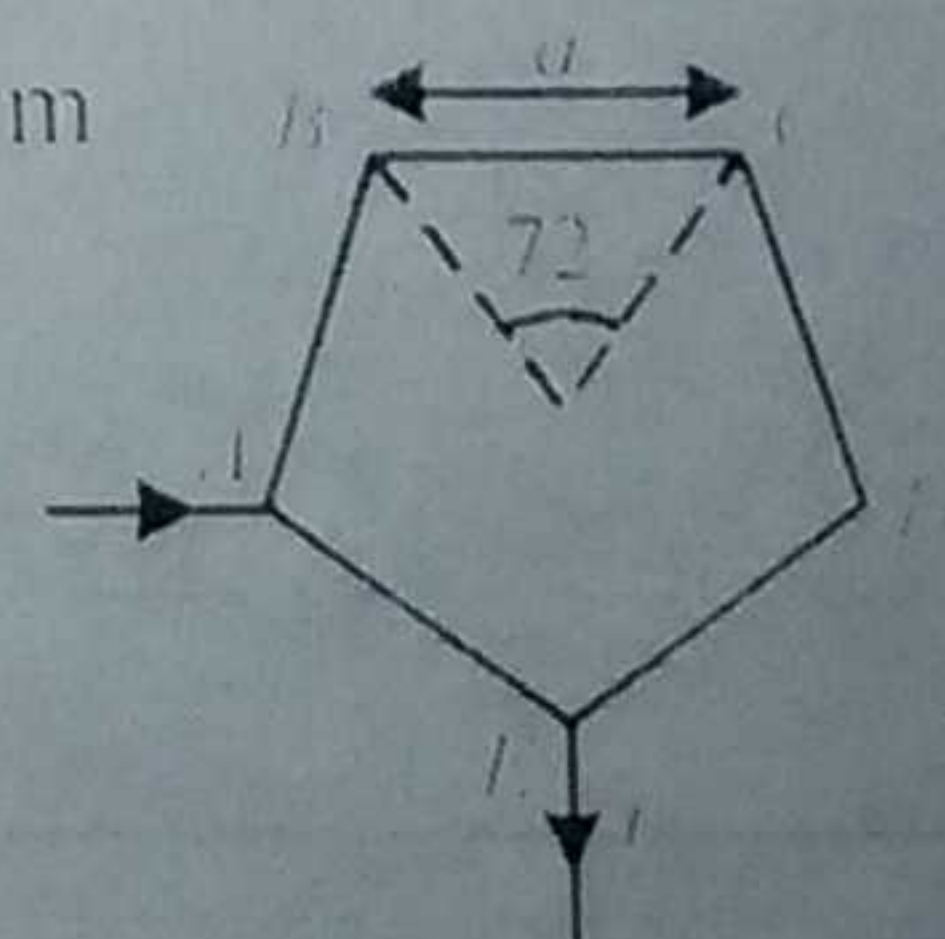
7. A moving charge produces :
 (A) electric field only (B) magnetic field only (C) both of them (D) None of these

8. Two parallel, long wires carry currents i_1 and i_2 with $i_1 > i_2$. When the current are in the same direction, the magnetic field at a point midway between the wire is $10 \mu T$. If the direction of i_2 is reversed, the field becomes $30 \mu T$. The ratio i_1/i_2 is :

(A) 4 (B) 3 (C) 2 (D) 1

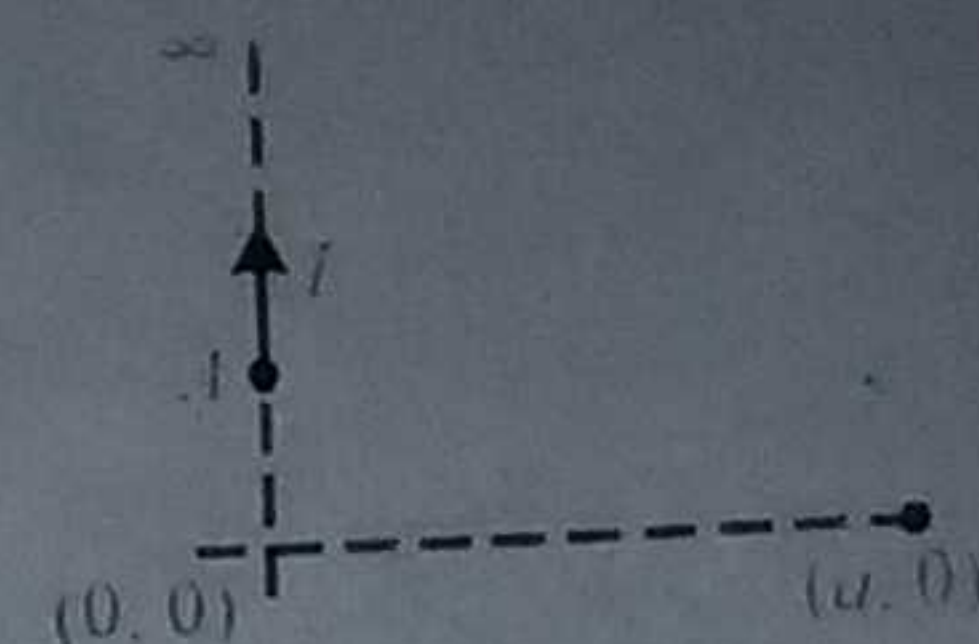
9. Magnetic field strength at the centre of regular pentagon made of a conducting wire of uniform cross section area as shown in figure is :

(A) $\frac{5\mu_0 i}{4\pi a} \left[2 \sin \frac{72^\circ}{2} \right]$ (B) 0
 (C) Not zero if current ' i ' leaves D point instead of E
 (D) Zero even if the current ' i ' leaves point D instead of point E



10. An infinitely long wire carrying current I is along y axis such that its one end is at point $A(0, b)$ while the wire extends upto $+\infty$. The magnitude of magnetic field strength at point $(a, 0)$:

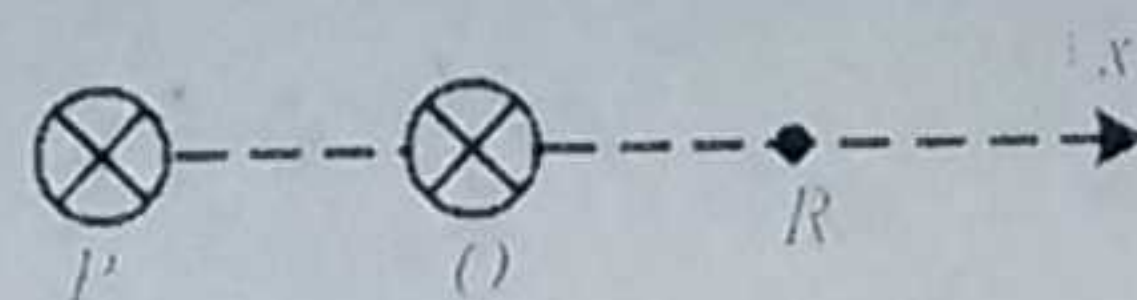
(A) $\frac{\mu_0 I}{4\pi a} \left(1 + \frac{b}{\sqrt{a^2 + b^2}} \right)$ (B) $\frac{\mu_0 I}{4\pi a} \left(1 - \frac{b}{\sqrt{a^2 + b^2}} \right)$
 (C) $\frac{\mu_0 I}{4\pi a} \left(\frac{b}{\sqrt{a^2 + b^2}} \right)$ (D) None of these



11. Consider a long, straight wire of cross-section area A carrying a current i . Let there be n free electrons per unit volume. An observer places himself on a trolley moving in the direction opposite to the current with a speed $v = (i/nAe)$ and separated from the wire by a distance r . The magnetic field seen by the observer is :

(A) $\frac{\mu_0 i}{2\pi r}$ (B) zero (C) $\frac{\mu_0 i}{\pi r}$ (D) $\frac{2\mu_0 i}{\pi r}$

12. Two long parallel wires carrying currents 2.5 amps and I amps in the same direction (directed into the plane of the paper) are held at P and Q respectively such that they are perpendicular to the plane of paper. The points P and Q are located at a distance of 5 m and 2 m respectively from a collinear point R .



- (a) An electron moving with a velocity of 4×10^5 m/s along the positive X -direction experiences a force of magnitude 3.2×10^{-20} N at the point R . Find the value of I .
 (b) Find all the positions at which a third long-parallel wire carrying a current of magnitude 2.5 amps may be placed so that the magnetic induction at R is zero.

13. A long, straight wire carries a current along the Z -axis. One can find two points in the X - Y plane such that :

- (A) the magnetic fields are equal.
 (B) the directions of the magnetic fields are the same.
 (C) the magnitudes of the magnetic fields are equal.
 (D) the field at one point is opposite to that at the other point.

14. A proton, a deuteron and an α -particle having the same kinetic energy are moving in circular trajectories in a constant magnetic field. If r_p , r_d and r_α denote respectively the radii of the trajectories of these particles then :

(A) $r_\alpha = r_p < r_d$ (B) $r_\alpha > r_d > r_p$ (C) $r_\alpha = r_d > r_p$ (D) $r_p = r_d = r_\alpha$

15. Three infinitely long thin wires, each carrying current i in the same direction, are in the x - y plane of a gravity free space. The central wire is along the y -axis while the other two are along $x = \pm d$.

- (a) Find the locus of the points for which the magnetic field B is zero.
 (b) If the central wire is displaced along the z -direction by a small amount and released, show that it will execute simple harmonic motion. If the linear density of the wires is λ , find the frequency of oscillation.

16. A hollow tube is carrying an electric current along its length distributed uniformly over its surface. The magnetic field :
 (A) increases linearly from the axis to the surface (B) is constant inside the tube
 (C) is zero at the axis (D) is zero just outside the tube

17. An α particle is accelerated by a potential difference of 10^4 V. Find the change in its direction of motion, if it enters normally in a region of thickness 0.1 m having transverse magnetic induction of 0.1 Tesla. (Given : mass of α -particle is equal to 6.4×10^{-27} kg)

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ANSWER KEY

1. (C)(D) 2. (C) 3. (A)(B)(C) 4. [12 cm] 5. [D - electron, B - α -particle] 6. [$M^3 L^{-2} T^4 Q^{-1}$] 7. (C)
 8. (C) 9. (A)(D) 10. (B) 11. (A) 12. [(a) 4.1. (b) (i) current directed into the plane of paper 1 m from R on RQ (away from Q) (ii) current directed out of from paper, 1 m from R on RQ (between R and Q)]

13. (B)(C)(D) 14. (A) 15. [(a) $x = 0$ and $x = \pm \frac{d}{\sqrt{3}}$ (b) $\frac{1}{2\pi d} \sqrt{\frac{\mu_0}{\pi \lambda}}$] 16. (B)(C) 17. [30°]